

TABLE 1
IMPORTANT LANGUAGE MODELS

Model	Release Date	Description
Falcon LLM	September 2023	This is a foundational large language model (LLM) with 180 billion parameters, trained on 3500 billion tokens.
Llama-2	July 2023	Meta AI developed LLaMA-2. It is an open and efficient foundation language model trained on trillions of tokens with publicly available datasets.
ChatGPT	November 2022	This model interacts in a conversational way. The dialogue format makes it possible for ChatGPT to answer follow up questions, admit its mistakes, challenge incorrect premises, and reject inappropriate requests.
Flan	October 2022	This fine-tuned language model is a collection of datasets used to improve model performance and generalisation to unseen tasks. 1800 tasks were phrased as instructions and used to fine tune the model.
Mistral 7B	September 2023	This is a pretrained generative text model with 7 billion parameters. The model is used in mathematics, code generation, and reasoning. It is suitable for real-time applications where quick responses are essential.
PaLM2	May 2023	This language model performs multilingual and reasoning capabilities.
Code Llama	August 2023	Designed for general code synthesis and understanding, it is a 34-billion parameter language model.
Phi-2	June 2023	Released by Microsoft Research Lab, this is a 2.7-billion parameter language model. Phi-2 can be prompted using a QA format, a chat format, and the code format.

Objectives and goals. The first step in the prompt engineering process is understanding the problem. Understanding the goal of the conversation and the objectives to be achieved are completed during this step. It addresses various problems like question-answering, text generation, sentiment analysis, etc. Analysis of tasks and constraints, and prediction of model's responses to various inputs are done.

Prompt creation. Initial prompt creation and its importance are covered during this step. Relevant data and meaningful keywords to train the model for generating the desired response/output are identified.

Response evaluation. During this step, effective use of the initial prompt is made. Clear and concise instructions are provided in the prompt. These prompts are kept simple and clear by using plain language. Identification of discrepancies, understanding limitations, and refining of prompts are done during this step.

Iterate and refine. Assessment of the model's output and its accuracy in terms of achieving the desired objective are done during this step. Any misalignment or flaws are captured as part of evaluation of the model's response. The nature of misalignment is identified, and the prompt instructions are adjusted and refined. The evaluation of the model's response is repeated to identify any remaining shortcomings. The process is iterated multiple times to refine the prompt effectively. This helps in improving the accuracy of the output.

Testing on different models. In this step, the performance of the generated prompts is evaluated and refined. A variety of test cases are developed to perform these tasks. Various large language models are selected and the refined prompt is applied to each model. The responses generated by each model are assessed, and the alignment between the responses and the desired objective is analysed. Finally, the processes are iterated by reapplying the refined

prompt to the models based on insights gained from testing.

Scale prompt. This is a process to extend the value of a prompt across more contexts, tasks, or levels of automation. Automating prompt generation saves a lot of time and can minimise human error. It helps in establishing uniformity in the prompt creation procedure. It also helps in achieving better customer experience and cost reduction.

Table 1 summarises some of the recent language models and how they successfully apply the latest and most advanced prompt engineering techniques.

Prompt engineering helps us achieve optimised outputs with minimum effort. Well defined prompts help the generative AI models to create accurate and personalised responses. Also, prompt engineering is likely to become increasingly integrated with other technologies, such as virtual assistants, chatbots, and voice-enabled devices. This will enable users to interact with technology more seamlessly and effectively, improving the overall user experience.

Prompt engineering is more an art than technology. The basic rule is that good prompts lead to good results. **EFY**

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